

To prove Boolean Algebra properties we can perform algebraic manipulations :

For example

$$\text{13a: } X + X \cdot Y = X$$

We work with left hand side of the property

$$X + X \cdot Y = X \text{ by using 6a}$$

$$x \cdot 1 = x$$

$$X \cdot 1 + X \cdot Y = X$$

$$X \cdot 1 + X \cdot Y = X \text{ by using 12a}$$

$$x \cdot (y + z) = x \cdot y + x \cdot z$$

$$X \cdot (1 + Y) = X$$

$$X \cdot (1 + Y) = X \text{ by using 10b}$$

$$x + y = y + x$$

$$X \cdot (Y + 1) = X$$

$$X \cdot (Y + 1) = X \text{ by using 5b}$$

$$x + 1 = 1$$

$$X \cdot 1 = X$$

$$X \cdot 1 = X \text{ by using 6a}$$

$$x \cdot 1 = x$$

$$X = X$$